

MATHIEU KUBIK
4th-year PhD student, Université Paris-Cité

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EDUCATION

PhD, Université Paris-Cité – Mathematics	2022–2026
<i>Advisor:</i> Emmanuel Letellier. <i>Title:</i> The weight filtration on the cohomology of character varieties	
MSc, Sorbonne Université – Pure Mathematics	2021–2022
Algebraic geometry, algebraic topology. <i>Grade:</i> 17.93/20	
MSc, Columbia University – Operations Research	2020–2021
Convex optimization, stochastic analysis. GPA: 4.09/4.0	
Engineering degree, École Polytechnique	2017–2021
Mathematics and Computer Science. <i>Graduation rank:</i> 10/540	
Preparatory classes, Lycée Pierre de Fermat	2015–2017

PUBLICATIONS

The simultaneous semi-random model for TSP, Eric Balkanski, Yuri Faenza, Mathieu Kubik, IPCO, 2022

TALKS

- 2024: *Counting indecomposable parabolic bundles*, Groupes, Représentations et Géométrie seminar (Université Paris-Cité)
- 2024: *La fibration de Hitchin*, PhD students seminar, IMJ-PRG (Sorbonne Université)
- 2024: *Moduli of parabolic bundles*, Arbeitsgemeinschaft Geometry and Representation Theory around the $P = W$ Conjecture (MFO, Oberwolfach)
- 2023: *Mixed Hodge theory*, RéGA seminar (Institut Henri-Poincaré)
- 2022: *Mixed Hodge theory*, $P = W$ working group (LMO, Orsay)

TEACHING

- 2025/2026: *Introduction to Python Programming, Web Programming* (TA). 192h
- 2024/2025: *Algebra 2* (L3 Math-CS), tutoring (L1–L3). 64h
- 2023/2024: *Elementary Mathematics* (L1 Chemistry), tutoring (L1–L3). 64h
- 2022/2023: *Elementary Mathematics* (L1 Chemistry), oral examinations (L1 Math). 64h

PROFESSIONAL EXPERIENCE

Teaching and Research Fellow (ATER), Université Paris-Cité	2025–2026
Teaching in undergraduate computer science	
Research internship, Polytechnique Montréal	2020
Constraint programming and reinforcement learning	
Data science internship, BCI Media Group	2019
Natural language processing	
Military internship, French National Gendarmerie	2017–2018
Operational support to the company commander	